Parallel & Distributed Computing Lab

IMSciences

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Week 2, Lab 1

Amdahl's Law

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1 Ideas

$$\text{speedup} = \frac{T_{\text{serial}}}{T_{\text{parallel}}}$$

How many times faster than **one** worker. If serial took 8 seconds and parallel took 2 seconds, speedup = 4. That is not the same as "saved 6 seconds".

2 Worked example

A program takes **20 seconds** on one core.

- Loop (can run in parallel): **16 s**
- Other work (stays serial: start, print, files): **4 s**

With $N = 1$ worker on the loop: still $16 + 4 = 20$ s. Speedup = 1. Extra cores on the machine do not help if the program still uses one.

With $N = 4$ workers on the loop:

$$T = 4 + \frac{16}{4} = 4 + 4 = 8 \text{ s}$$

$$\text{speedup} = \frac{20}{8} = 2.5$$

Not 4×. The 4 s of other work did not shrink.

If the loop became instant ($N \to \infty$), the wait is still 4 s. Best possible speedup:

$$\frac{20}{4} = 5$$

The serial remainder is a floor. Time cannot go below it.

3 Amdahl's law

P = fraction of time that *can* run in parallel. $1 - P$ stays serial. N = workers.

$$S = \frac{1}{(1 - P) + \dfrac{P}{N}}$$

Same example: $P = 16/20 = 0.8$, so $1 - P = 0.2$, $N = 4$:

$$S = \frac{1}{0.20 + 0.80/4} = \frac{1}{0.20 + 0.20} = \frac{1}{0.40} = 2.5$$

The parentheses are required. Without them the value is wrong.

4 Time view (same numbers)

$$T_{\text{parallel}} = T_{\text{serial_part}} + \frac{T_{\text{parallel_part}}}{N}$$

$$8 = 4 + \frac{16}{4}$$

The formula and the time view must give the same result.

5 Numbers for this example

$N = 4$, total 20 s, loop 16 s, other 4 s:

- $P = 0.8$
- $T_{\text{parallel}} = 8$ s
- speedup $= 2.5$
- floor (loop $\to 0$) $= 4$ s, max speedup $= 5$